Linear forms, moment determinants, and arithmetic constraints for even beta values

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Abstract

This supplement studies arithmetic constraints in attempts to prove irrationality of an individual even Dirichlet beta value or of a member of a small explicit set. For a positive moment construction involving $\beta(4)$ and $\zeta(4)$, we determine the exact power of two in every principal minor of its beta-coefficient matrix. The resulting necessary denominator growth rules out a specified certificate based on one-sided rational approximation to $\zeta(4)$. We also give controlled rational elimination of the unwanted period, uniform divisibility results for compressed determinants, and a two-sided estimate for the residual decay of a thirteen-value Gram construction. A separate linear-form construction provides eleven independent, uniformly small integer forms in fourteen candidate beta values, with nonvanishing still unresolved. These results identify usable mechanisms and precise remaining obstacles; they do not establish a new irrationality theorem.

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1 What a small determinant must accomplish

For $s > 1$, the Dirichlet beta function is

$$\beta(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^s}.$$

Its values at positive odd integers are nonzero rational multiples of powers of π. For even arguments, explicit irrationality results remain much harder to obtain. Our earlier work proves that at least one of $\beta(4), \beta(6), \ldots, \beta(32)$ is irrational [10], refining the construction of Krisman [5]. The investigations here sought a smaller set, or the individual value $\beta(4)$.

The recurring difficulty is easy to state. A rational-coefficient polynomial can be very small at the target constants because all of its coefficients share a very small rational factor. That factor has no bearing on irrationality. What matters is the value after the coefficients have been made relatively prime integers. Positivity can prove that this value is nonzero, but cannot make its arithmetic cost disappear.

1.1 The exact denominator result

Our main result concerns a positive moment functional introduced by Carreira [2, Section 11]. For a positive integer K, interlacing K integer and K half-integer poles gives matrices whose entries are rational combinations of 1, $\beta(4)$, and $\zeta(4)$. One possible strategy is to retain a subspace on which the $\zeta(4)$ coefficient is positive, then replace $\zeta(4)$ by an upper rational approximation. Positivity supplies nonvanishing automatically. The question becomes whether the resulting determinant remains small after clearing its denominators.

Sections 2–4 answer that question for a precise certificate. Theorem 3.1 determines the exact dyadic valuation of every principal minor of the beta-coefficient matrix. On the positive subspace under consideration it forces a denominator of size at least

$$\exp\bigl((\log 2)K^2 - O(K \log K)\bigr).$$

This is a necessary cost of the whole polynomial, rather than the cost of a convenient way of clearing its entries. Theorem 4.2 shows that the usual error bound q^{-2} for a rational approximation cannot pay that cost in the specified certificate, for any choice of its denominator q. The distinction between this certificate and the actual approximants is essential and is made explicit there.

The arithmetic mechanism may be useful separately from this application. A finite binomial average replaces a residue sum with substantial cancellation. In its Krawtchouk basis, multiplication has a triangular reduction modulo two. Known orthogonal norms then account for the entire determinant valuation. All properties of that basis used here are derived from its generating function.

1.2 Other constructions and what they establish

There is another way to remove $\zeta(4)$: restrict the coefficient quadratic form to a rational subspace on which it vanishes. Section 5 constructs such subspaces with controlled height. A local factorial cancellation reduces an apparently exponential vector-height cost to a polynomial one. Meyer and Cassels provide the classical existence and small-zero inputs; the local cancellation and integral lift supply the quantitative adaptation.

Section 6 develops arithmetic savings for the version that keeps only the added low-degree directions. It combines pole collisions, exact low-rank terms, and dyadic Cauchy-minor divisibility.

The resulting upper bound for primitive arithmetic remains insufficient for the available decay comparison. An insufficient upper bound does not prove that the actual primitive values fail.

Section 7 treats a different positive Gram system involving $\beta(4), \dots, \beta(28)$. Removing a rational block leaves a seven-by-seven determinant. We prove that its logarithm has linear, rather than quadratic, scale in the family index. Its primitive arithmetic remains unresolved. Section 8 returns to linear forms: formal rank and integral removal of zero multipliers lead to eleven controlled independent directions, but do not establish nonvanishing at the actual constants.

These are later investigations than those collected in [8, 9]. Positive determinant criteria and their arithmetic are established methods [11]; our contribution lies in the particular reductions, constructions, and estimates proved here. The separate constructions are presented as case studies, each with its conclusion and missing input. A reader principally interested in the exact valuation and its application can read Sections 2–4 independently. The prerequisites are elementary p-adic valuations, Gram and Schur-complement linear algebra, and lattice reduction. The finite orthogonal-polynomial identities needed for the main theorem are derived below.

1.3 Primitive normalization and growing degree

For a nonzero rational polynomial D, let $q_D > 0$ be the unique rational number such that $q_D D$ has integer coefficients with gcd one. We retain the sign of D; whenever D is a positive Gram determinant, its primitive value is positive. A nonzero rational factor in D has no separate effect on this primitive polynomial, apart from sign.

If D has degree at most h and a target number equals a/b, then $b^h q_D D(a/b)$ is an integer. For fixed h, a positive primitive value tending to zero gives an irrationality contradiction. When $h = O(K)$, one must also beat every fixed factor b^h; a bound $q_D D(\beta(4)) \le \exp(-\varepsilon K^2)$ suffices. We will keep this degree cost visible in each comparison.

For a univariate determinant with leading nonzero coefficient ℓ_D, it is often useful to separate the invariant quantities

$$A_D = q_D|\ell_D|, \qquad \delta_D = \frac{D(\beta(4))}{|\ell_D|}, \qquad q_D D(\beta(4)) = A_D \delta_D. \tag{1.1}$$

Here A_D is the absolute leading coefficient of the primitive polynomial. Both quantities are unchanged by multiplying D by a positive rational number. This prevents an arbitrary matrix scale from being counted as either arithmetic improvement or analytic decay.

2 One positive functional, two periods

To put both periods in the same positive matrix, use Carreira's functional [2, Definition 24 and Lemma 25], before restricting its resolvent to half-integers. Set

$$g(y) = \frac{1}{\sinh(\pi y)}, \qquad w(y) = -\frac{y^4}{6} g'''(y), \qquad \nu(f) = \int_0^\infty f(4y^2) w(y)\, \mathrm{d}y. \tag{2.1}$$

The series $g(y) = 2\sum_{j\ge 0} e^{-(2j+1)\pi y}$ shows that $w > 0$. The weight is bounded near zero and decays exponentially times a polynomial at infinity. All the integrals below converge.

2.1 The resolvent, including its rational correction

For every $a > 0$, the inherited identity is

$$\int_0^\infty \frac{w(y)}{y^2+a^2}\,\mathrm{d}y = a^3 \sum_{k\ge 0} \frac{(-1)^k}{(k+a)^4} - \frac{1}{2a}. \tag{2.2}$$

We include its proof because the final rational correction is important in the later prime calculations. Put $f(y) = y^4/(y^2+a^2)$. Three integrations by parts, using $f'''(y) = 6a^3 \operatorname{Im}(a-iy)^{-4}$, give the left side as $a^3 \int_0^\infty \operatorname{Im}(a-iy)^{-4} g(y)\,\mathrm{d}y$. At zero the three boundary terms are $O(y)$; at infinity they vanish exponentially. Now use

$$\operatorname{Im}(a-iy)^{-4} = \frac{1}{6}\int_0^\infty x^3 e^{-ax}\sin(xy)\,\mathrm{d}x, \qquad \int_0^\infty \sin(xy)g(y)\,\mathrm{d}y = \frac{1}{2}\tanh(x/2).$$

The second formula follows from the exponential series for g and the logarithmic derivative of Euler's product for $\cosh(x/2)$. The double integral is absolutely convergent, using $|\sin(xy)| \le xy$ near zero. Expanding $\tanh(x/2) = 1 + 2\sum_{k\ge1}(-1)^k e^{-kx}$ and integrating proves (2.2).

Write X and Z for indeterminates representing $\beta(4)$ and $\zeta(4)$. For odd $l = 2j+1$ and for $m \ge 1$, define

$$b_l = \frac{(-1)^j l^3}{2}, \qquad r_l^o = -b_l \sum_{v=0}^{j-1} \frac{(-1)^v}{(2v+1)^4} - \frac{1}{4l}, \tag{2.3}$$

$$c_m = \frac{7(-1)^{m-1}m^3}{32}, \qquad r_m^e = -\frac{(-1)^{m-1}m^3}{4} \sum_{v=1}^{m-1} \frac{(-1)^{v-1}}{v^4} - \frac{1}{8m}. \tag{2.4}$$

Empty sums are zero. The change of variable in (2.1) and (2.2) give the formal evaluations

$$\nu\big((u+l^2)^{-1}\big) = b_l X + r_l^o, \qquad \nu\big((u+4m^2)^{-1}\big) = c_m Z + r_m^e. \tag{2.5}$$

They become genuine integral identities at the actual periods.

2.2 A fixed cofactor normalization

For $K \ge 1$, put

$$a_m = 4m^2, \qquad J(u) = \prod_{m=1}^{K}(u+a_m), \qquad O(u) = \prod_{j=0}^{K-1}(u+(2j+1)^2),$$
$$D(u) = J(u)O(u), \qquad J_m(u) = \frac{J(u)}{u+a_m}.$$

Here O is a polynomial, not an order symbol. Define the symmetric pencil

$$M_K(X,Z) = R_K + XB_K + ZC_K, \qquad (M_K)_{mn} = \nu(J_m J_n/D). \tag{2.6}$$

The right side is evaluated formally by (2.5). At $(X,Z) = (\beta(4), \zeta(4))$ it is a positive definite Gram matrix: the functions $J_m/\sqrt{D}$ are independent and the measure is positive.

The complete coefficients have a particularly useful form. With l running over the odd integers from 1 to $2K-1$, set

$$e_m = \frac{J'(-a_m)}{O(-a_m)} = -\frac{4^{2K-1}}{2m^2\binom{2K-2m}{K-m}\binom{2K+2m}{K+m}}, \tag{2.7}$$
$$\rho_l = \frac{J(-l^2)}{O'(-l^2)}, \qquad V_{ml} = \frac{1}{a_m - l^2}.$$

Partial fractions give

$$C_K = \operatorname{diag}(e_m c_m), \quad B_K = V \operatorname{diag}(\rho_l b_l) V^T, \quad R_K = \operatorname{diag}(e_m r_m^e) + V \operatorname{diag}(\rho_l r_l^o) V^T. \tag{2.8}$$

An even pole contributes only to its corresponding diagonal entry; an odd pole contributes the rank-one residue $\rho_l V_{\cdot l} V_{\cdot l}^T$. There is no polynomial quotient here, since the numerator degree is $2K-2$ and the denominator degree is $2K$.

For even m,

$$(C_K)_{mm} = \frac{7\, 2^{4K-8} m}{\binom{2K-2m}{K-m}\binom{2K+2m}{K+m}} > 0. \tag{2.9}$$

Thus the even-coordinate principal subspace permits an upper rational substitution for $\zeta(4)$ without losing positivity. We will first determine its arithmetic exactly.

3 An exact dyadic determinant formula

Individual residues in (2.8) can have large dyadic denominators. Adding their valuations cannot determine the denominator of their sum. We instead express the whole coefficient matrix through one finite orthogonal system. This identifies the cancellation exactly.

3.1 Replacing residues by a finite binomial average

Let $N = 2K-1$ and let L be the sum of N independent variables taking the values 1 and -1 with equal weights. Expectations below are finite rational sums. At an odd point $l = 2j+1$,

$$O'(-l^2) = \frac{(-1)^j 4^{K-1}(K-1-j)!(K+j)!}{l}, \qquad \Pr(L=l) = \frac{N!}{2^N (K-1-j)!(K+j)!}.$$

Pairing l and $-l$ gives

$$(B_K)_{mn} = \frac{1}{2N!}\mathbb{E}\left[\frac{L^4 J(-L^2)}{(a_m - L^2)(a_n - L^2)}\right]. \tag{3.1}$$

Every denominator inside the expectation is odd and nonzero. Off the diagonal the two denominator factors cancel factors of J; on the diagonal a rational remainder survives. The argument below handles both together.

3.2 The finite orthogonal basis

Define the monic Krawtchouk polynomials by

$$\sum_{n\ge 0} p_n(L)\frac{z^n}{n!} = (1+z)^{(N+L)/2}(1-z)^{(N-L)/2}. \tag{3.2}$$

At each support point, the right side is the product of N factors $1+z\varepsilon_i$. The expectation of the product of two such generating functions is $(1+zs)^N$. Comparing coefficients and then differentiating (3.2) gives

$$\mathbb{E}[p_n p_r] = \begin{cases} n!N!/(N-n)!, & n = r \le N, \\ 0, & n \ne r, \end{cases} \tag{3.3}$$

$$L p_n = p_{n+1} + n(N-n+1)p_{n-1}, \qquad p_0 = 1,\ p_1 = L. \tag{3.4}$$

In particular p_n is integral and has the parity of n. Put $t = L^2$ and $q_j(t) = p_{2j}(L)$. The K monic polynomials $q_0, \dots, q_{K-1}$ are orthogonal, with norms

$$h_j = (2j)!N!/(N-2j)!. \tag{3.5}$$

There are exactly K distinct support points for t. The polynomial $q_K = p_{N+1}$ vanishes at all of them, since the evaluated generating function has degree N. Multiplication by t is therefore represented exactly by a finite K-by-K matrix T in this basis.

Applying (3.4) twice gives

$$\begin{aligned} tq_j &= q_{j+1} + \tau_j q_j + u_j q_{j-1}, \\ \tau_j &= (4j+1)N - 8j^2, \qquad u_j = (2j)(2j-1)(N-2j+1)(N-2j+2). \end{aligned} \tag{3.6}$$

Terms outside the basis are omitted on the support. Every τ_j is odd and every u_j is divisible by four. Thus T is triangular modulo two with diagonal one. This supplies more than invertibility: it will make every leading principal block of the relevant multiplier a unit.

3.3 Valuation and nonvanishing for every subset

For nonzero rational r, let $v_2(r)$ denote its exponent of two. Write $\mathbb{Z}_{(2)}$ for the rationals with odd denominator. An odd determinant is a unit in this ring.

Theorem 3.1. *For every nonempty $S \subseteq \{1, \dots, K\}$, with $d = |S|$, set $\Delta_S = \prod_{m<n,\, m,n\in S}(4n^2-4m^2)$. Then $\det(B_K)_S \neq 0$ and*

$$v_2\big(\det(B_K)_S\big) = 2v_2(\Delta_S) - d + \sum_{j=0}^{d-1} \left(v_2((2j)!) - v_2((2K-1-2j)!)\right). \tag{3.7}$$

Proof. The purpose of the first change of basis is to replace the rational functions in (3.1) by polynomials of degree below d. Put

$$W_S(t) = \prod_{m\in S}(a_m - t), \qquad F_S(t) = \frac{t^2 J(-t)}{W_S(t)^2}, \qquad r_m(t) = \frac{W_S(t)}{a_m - t}.$$

Then $(B_K)_S$ is $(2N!)^{-1}$ times the Gram matrix $\mathbb{E}[F_S r_m r_n]$. The coefficient determinant of the r_m in the monomial basis is $\pm\Delta_S$: evaluate at the a_m to obtain a diagonal matrix and divide by the evaluation Vandermonde. Passing to the monic basis $q_0, \dots, q_{d-1}$ changes no further determinant.

Every $a_m I - T$ is invertible over $\mathbb{Z}_{(2)}$. Consequently

$$F_S(T) = T^2 J(-T) W_S(T)^{-2}$$

is defined over that ring and reduces to T^{K+2-2d} modulo two. Negative exponents are allowed because T is invertible. This reduction is triangular with diagonal one, so its leading d-by-d block has odd determinant.

The functional calculus is finite: multiplication by t acts on the K support points and W_S vanishes at none of them. Orthogonality therefore gives

$$\big(\mathbb{E}[q_i F_S q_j]\big)_{i,j<d} = \operatorname{diag}(h_0, \dots, h_{d-1})[F_S(T)]_{i,j<d}.$$

The last determinant is a unit, hence nonzero. Including the basis factor and the scalar $(2N!)^{-d}$ gives $2v_2(\Delta_S) - d(1 + v_2(N!)) + \sum_{j<d} v_2(h_j)$. Substitution of (3.5) proves the formula. □

For example, $K = 2$ and $S = \{2\}$ give $B_{22} = -167/140$ and $C_{22} = 1/5$. The theorem gives $v_2(B_{22}) = -2$. After making the pencil monic in Z, its X coefficient is $-167/28$, again of valuation -2. The example illustrates why the coefficient normalization must accompany the matrix calculation.

3.4 The denominator forced by the positive restriction

From now on let

$$S_K = \{2, 4, \dots, 2d\}, \qquad d = \lfloor K/2 \rfloor, \quad K \geq 2, \tag{3.8}$$

and define the polynomial monic in Z and its integer clearer by

$$P_K(X, Z) = \det(C_S^{-1}(R_S + XB_S) + ZI_d), \qquad A_K = \min\{A \in \mathbb{Z}_{>0} : AP_K \in \mathbb{Z}[X, Z]\}. \tag{3.9}$$

Since P_K is monic, $A_K P_K$ is primitive. Its nonzero coefficient of X^d is $\det(C_S^{-1}B_S)$, so the reduced denominator of this one coefficient must divide A_K.

Write $s_2(n)$ for the binary digit sum. The elementary factorial formula $v_2(n!) = n - s_2(n)$ gives $v_2\binom{2n}{n} = s_2(n)$. The selected Vandermonde is

$$\Delta_S = 2^{2d(d-1)} \prod_{j=1}^{d} \frac{(2j-1)!}{j}, \qquad v_2(\Delta_S) = 3d^2 + O(d \log(d+1)).$$

Theorem 3.1 and (2.9) now yield

$$\begin{aligned} v_2(\det B_S) &= 8d^2 - 2Kd + O(K \log K), \\ v_2(C_{mm}) &= 4K - 8 + v_2(m) - s_2(K-m) - s_2(K+m), \\ v_2(\det(C_S^{-1}B_S)) &= 8d^2 - 6Kd + O(K \log K) = -K^2 + O(K \log K). \end{aligned}$$

We have proved the necessary bound

$$\log A_K \geq (\log 2)K^2 - O(K \log K). \tag{3.10}$$

The restriction matters. On the full coordinate space there is instead the identity

$$\det(C_K^{-1}B_K) = (-1)^K (2/7)^K \binom{2K}{K}^3. \tag{3.11}$$

Indeed $J/O = 1 + \sum_l \rho_l/(u + l^2)$ implies, by divided differences at $u = -a_m$ and the diagonal derivative, $V \operatorname{diag}(\rho_l) V^T = -\operatorname{diag}(e_m)$. Taking determinants in (2.8) gives $(-1)^K \prod_l b_l / \prod_m c_m$, which simplifies to (3.11). That full-space coefficient has no dyadic denominator, but C_K is indefinite. Thus the necessary cost in (3.10) belongs to the particular positive restriction, not to every use of the functional.

4 A limitation of one-sided zeta substitution

An upper rational approximation to $\zeta(4)$ increases a positive $\zeta(4)$-coefficient matrix. It therefore solves the nonvanishing problem without a perturbation or conditioning estimate. We now compare this advantage with the necessary arithmetic cost just proved.

4.1 A lower bound for the actual determinant

Let $S = S_K$ and normalize the actual positive matrix as

$$T_K = C_S^{-1/2} M_K(\beta(4), \zeta(4))_S C_S^{-1/2} > 0.$$

The bounds $4^n/(2n+1) \le \binom{2n}{n} \le 4^n$ in (2.9) give

$$\frac{7m}{256} \le C_{mm} \le \frac{7m(2K-2m+1)(2K+2m+1)}{256}.$$

Hence $|\log\det C_S| = O(K\log K)$; it does not change a quadratic logarithmic rate.

Lemma 4.1. *With*

$$D_* = -\log 2 + \tfrac{1}{2}\log(5/2) + \arctan(1/2) + \tfrac{1}{4}\arctan 2 + \tfrac{\pi}{8}, \tag{4.1}$$

one has $\log\det T_K \ge -D_* K^2 + o(K^2)$. *Here* $D_* = 0.8981320555\ldots$.

Proof. Only one fixed integration region is needed. If $t = \pi y$ and $r = e^{-2t}$, direct differentiation gives

$$w(y)e^t = \frac{t^4}{3\pi}\frac{1+23r+23r^2+r^3}{(1-r)^4} \ge \frac{1}{48\pi},$$

using $1 - r \le 2t$. Also $J(4y^2)/O(4y^2) \ge 1$. With $m_i = 2i$,

$$(M_S)_{ij} = \frac{1}{16}\int_0^\infty \frac{J(4y^2)}{O(4y^2)}\frac{w(y)}{(y^2+m_i^2)(y^2+m_j^2)}\,\mathrm{d}y.$$

Expanding two determinants and applying Fubini (Andréief's identity) expresses $\det M_S$ as $16^{-d}/d!$ times an integral of the squared Cauchy determinant and the d weights. The squared determinant is

$$\frac{\prod_{i<j}(m_j^2-m_i^2)^2(y_j^2-y_i^2)^2}{\prod_{i,j}(m_i^2+y_j^2)^2}. \tag{4.2}$$

Restrict the positive integral to the ordered boxes

$$\frac{sK(i-3/4)}{d} \le y_i \le \frac{sK(i-1/4)}{d}, \qquad 1 \le i \le d, \qquad s = \tfrac{1}{2}.$$

Their volume and all scalar prefactors cost only $O(K\log K)$ in the logarithm. The scaled pole and free coordinates approach uniform distributions on $[0,1]$ and $[0,s]$ respectively, with mass $\alpha = d/K \longrightarrow 1/2$.

The logarithmic sums converge uniformly throughout these boxes. Here are details at the singularities, where ordinary continuity does not suffice. Scaled successive coordinates have separation at least c/d, and the ith coordinate is at least ci/d. A strip $|i-j| \le \varepsilon d$ for the difference kernel contributes at most

$$\frac{C}{d}\sum_{h\le\varepsilon d}\log(Cd/h) = O(\varepsilon(1+|\log\varepsilon|)) + o(1)$$

after division by d^2. At the origin use the lower bounds $c(i+j)/d$ for a sum and $c_s(i^2+j^2)/d^2$ for the mixed squared sum. The corner contribution is $O(\varepsilon^2(1+|\log\varepsilon|)) + o(1)$. On the remaining compact regions the kernels are uniformly continuous. First letting d grow and then ε decrease proves the required convergence, including the missing Vandermonde diagonals.

The elementary integrals are

$$\begin{aligned} I &= \int_{[0,1]^2} \log|x^2 - v^2| \,\mathrm{d}x\,\mathrm{d}v = 2\log 2 - 3, \\ J(s) &= \int_{[0,1]^2} \log(x^2 + s^2 v^2)\,\mathrm{d}x\,\mathrm{d}v \\ &= \log(1+s^2) - 3 + \frac{\arctan s}{s} + s\arctan(1/s). \end{aligned}$$

For I, split into $\log|x-v|$ and $\log(x+v)$; for J, integrate in each variable. Formula (4.2) and the weight bound give

$$\liminf K^{-2}\log\det M_S \geq \alpha^2\{2I + 2\log s - 2J(s)\} - \frac{\pi\alpha s}{2}.$$

The powers of K in the products cancel up to $O(K\log K)$. Substituting $\alpha = s = 1/2$ gives $-D_*$. The normalization by $\det C_S$ has lower-order logarithm, as observed above. □

4.2 The certificate and its obstruction

Suppose $0 < p/q - \zeta(4) \leq q^{-2}$ and write $\epsilon = p/q - \zeta(4)$. At the actual beta value the normalized matrix becomes $T_K + \epsilon I$, which remains positive. Under the hypothetical rationality $\beta(4) = a/b$, the quantity

$$A_K b^d q^d P_K(a/b, p/q)$$

is a positive integer: each variable of P_K has degree at most d. Using only $\epsilon \leq q^{-2}$ gives the upper certificate

$$\mathcal{E}_K(q;b) = A_K b^d \det(qT_K + q^{-1}I). \tag{4.3}$$

The global polynomial clearer A_K is chosen before specializing X. To obtain the proposed contradiction, this certificate would have to tend to zero.

Theorem 4.2. *For every fixed positive integer b,*

$$\liminf_{K\to\infty} \frac{1}{K^2}\log\inf_{q>0}\mathcal{E}_K(q;b) \geq \gamma = \log 2 - D_*/2 = 0.2440811527\ldots > 0. \tag{4.4}$$

Thus the certificate (4.3) cannot tend to zero for any choice of positive $q = q_K$.

Proof. For every eigenvalue $\lambda_i > 0$ of T_K, $q\lambda_i + q^{-1} \geq 2\sqrt{\lambda_i}$. Hence

$$\det(qT_K + q^{-1}I) \geq 2^d\sqrt{\det T_K}$$

uniformly in $q > 0$. Combine this with (3.10) and Lemma 4.1. The factor b^d has logarithm $O_b(K)$. The positive sign does not require a numerical certificate: $\log 2 > 2/3$, $\log(5/2) < 1$, $\arctan(1/2) < 1/2$, $\arctan 2 < \pi/2$ and $\pi < 22/7$ give $D_* < 47/42$ and $\gamma > 3/28$. □

This conclusion concerns the displayed upper certificate. A lower bound for an upper bound is not a lower bound for the actual integer candidate. Exceptionally accurate approximations, cancellation after specializing X to a hypothetical rational value, and different positive subspaces are not excluded. The theorem closes the specified global-monic, standard-error argument; it does not close the irrationality problem for $\beta(4)$.

5 Eliminating a period at controlled height

Instead of approximating $\zeta(4)$, we can cancel its coefficient exactly. For a symmetric pencil $R + XB + ZC$, restriction by a rational matrix U eliminates Z when $U^T C U = 0$. Such a subspace is called totally isotropic for C. If U has independent columns, the actual compressed Gram matrix remains positive definite. The issue is constructing enough directions without paying an excessive height cost.

5.1 A larger cofactor space and the local cancellation

For this construction use the full basis

$$E_l(u) = \frac{D(u)}{u + l^2}, \quad 1 \le l \le 2K, \qquad \mathcal{G}_{lk}(X, Z) = \nu(E_l E_k / D).$$

It is important to include polynomial parts in this matrix. For $l \neq k$, $E_l E_k / D$ is an integer polynomial. On the diagonal, monic division gives

$$E_l^2 / D = Q_l(u) + D'(-l^2)/(u + l^2), \qquad Q_l \in \mathbb{Z}[u], \tag{5.1}$$

where

$$D'(-l^2) = \frac{(-1)^{l-1}(2K - l)!(2K + l)!}{2l^2}. \tag{5.2}$$

Integration by parts in (2.1) gives

$$\nu(u^e) = \tfrac{1}{4}\binom{2e+4}{3} T_{2e+1}, \qquad \tan z = \sum_{e \ge 0} T_{2e+1} \frac{z^{2e+1}}{(2e+1)!}. \tag{5.3}$$

For example the first moments are $1, 10, 224$. One derivation expands g in exponentials after transferring three derivatives to y^{2e+4}; the resulting odd-power sum is the classical tangent-number formula. The tangent numbers are integers, also directly from $(\tan z)' = 1 + \tan^2 z$. Thus all polynomial moments have denominator dividing four. Equations (2.5) and (5.1) show that $32\,\mathrm{lcm}(1, \dots, 2K)^4$ clears every coefficient of $\mathcal{G}$.

Both period-coefficient matrices of $\mathcal{G}$ are diagonal. The Z coefficient vanishes exactly on the K odd coordinates, and on the even coordinates it equals

$$\mathcal{C}_{2m,2m} = \frac{7}{256}(-1)^m m (2K - 2m)!(2K + 2m)!. \tag{5.4}$$

To choose an isotropic direction, first prescribe polynomial values rather than cofactor coefficients. For $\deg P < 2K$, put $\xi_m = m^2 P(-4m^2)$. Partial fractions and (5.2) give

$$[Z]\nu(P^2/D) = \frac{7}{4(4K)!} \sum_{m=1}^{K} (-1)^m m \binom{4K}{2K - 2m} \xi_m^2.$$

We will construct a zero of this quadratic form on each five-node block, then lift a proportional evaluation vector to an integral polynomial. Alternating signs permit cancellation, but real indefiniteness alone does not ensure a rational zero. We use the classical Meyer theorem in dimension five, followed by Cassels' quantitative small-zero bound [4, Introduction and Theorem 1.1].

Take five consecutive indices $a \le m \le b = a + 4 \le K$ and set

$$B_m^{\mathrm{loc}} = \frac{(2K - 2a)!}{(2K - 2m)!} \frac{(2K + 2b)!}{(2K + 2m)!}. \tag{5.5}$$

These are products of exactly eight positive integers at most $4K$. The integral diagonal form

$$\sum_{m=a}^{b} (-1)^m m B_m^{\mathrm{loc}} z_m^2$$

is nonsingular and indefinite. Meyer supplies a rational zero, and Cassels supplies a nonzero integer zero satisfying

$$\max_m |z_m| \le [15K(4K)^8]^2 =: Z_K. \tag{5.6}$$

The useful step is removal of a common factor before applying this bound. In evaluation coordinates the coefficients are proportional to $(-1)^m m\binom{4K}{2K-2m}$. Within the block their common positive rational factor is $(4K)!/((2K-2a)!(2K+2b)!)$. Dividing by it leaves precisely the polynomial-size coefficients in (5.5). An estimate using the binomial coefficients themselves would miss this saving.

For illustration, at $K=5$ the primitive equation is

$$-25194z_1^2 + 15504z_2^2 - 2907z_3^2 + 152z_4^2 - z_5^2 = 0.$$

The vector $(-42, 57, 136, 561, 0)$ is an exact zero. The general existence and height statement comes from Meyer and Cassels, not from extrapolating this example.

5.2 An integral lift and its determinant cost

Lift a block zero to the polynomial

$$P_z(u) = \sum_{m=a}^{b} z_m B_m^{\mathrm{loc}} E_{2m}(u). \tag{5.7}$$

It vanishes at every odd node and at every even node outside its block. At an even node in the block, (5.2) gives

$$m^2 P_z(-4m^2) = -\tfrac{1}{8}(2K-2a)!(2K+2b)!z_m.$$

Equivalently, substituting (5.4) gives its Z coefficient as $\frac{7}{256}(2K-2a)!(2K+2b)!\sum_m (-1)^m m B_m^{\mathrm{loc}} z_m^2 = 0$. The cofactor coefficients are integers bounded by

$$W_K^{\mathrm{full}} = 225K^2(4K)^{24}. \tag{5.8}$$

Theorem 5.1. *For every $K \ge 5$, there is an integral cofactor-coordinate matrix $\mathcal{U}$ of rank $h = K + \lfloor K/5 \rfloor$ such that $\mathcal{U}^T\mathcal{C}\mathcal{U} = 0$ and*

$$\det(\mathcal{U}^T\mathcal{U}) \le \left(5(W_K^{\mathrm{full}})^2\right)^{\lfloor K/5 \rfloor}.$$

Its compressed matrix is positive definite at $\beta(4)$ and has only rational and $\beta(4)$ entries. Its determinant polynomial has degree exactly K.

Proof. Adjoin the K odd coordinate vectors to one lift (5.7) from each of the disjoint blocks $[5i+1, 5i+5]$. Their supports are disjoint. They are independent, mutually $\mathcal{C}$-orthogonal, and isotropic. The volume bound is the product of their squared lengths. Positivity follows by restriction of the actual Gram matrix. The beta-coefficient matrix is nonsingular on the odd coordinates and zero on the added coordinates, so its rank is exactly K. The degree assertion also follows from the block factorization below. □

This is a height statement in the declared cofactor basis. The large monomial coefficients of the cofactors have not been bounded by a power of K. To see the precise transfer, let I range over the possibly nonzero row minors of $\mathcal{U}$: all odd coordinates and one even coordinate from each block. Cauchy–Binet and Gram Cauchy–Schwarz give

$$0 < \det(\mathcal{U}^T \mathcal{G} \mathcal{U})(\beta(4)) \le \max_I \det \mathcal{G}_{I,I}(\beta(4), \zeta(4)) \big(25 (W_K^{\mathrm{full}})^2\big)^{\lfloor K/5 \rfloor}. \tag{5.9}$$

Indeed the sum of the absolute row minors is the product of the five-entry ℓ^1 norms. The logarithm of its square is $O(K \log K)$. If a multiplier clears every coefficient of the same mixed minors $\det \mathcal{G}_{I,J}(X, Z)$, it clears the compressed polynomial too. Thus the projection adds no leading K^2 height cost to a comparison made in this same basis. It does not by itself supply that comparison.

For every fixed odd block length $L \ge 5$, the extension in Appendix A retains $(L-3)/2$ independent isotropic directions per block. The dimension becomes $K + (L-3)\lfloor K/L \rfloor/2$, still with $O_L(K \log K)$ logarithmic height cost. Fixing L before $K \to \infty$ is essential.

5.3 The rational block and what survives it

Order the odd coordinates first. The compressed matrix has the form

$$\begin{pmatrix} L_0 + X B_0 & T_0 \\ T_0^T & E_0 \end{pmatrix}, \qquad \det = \det E_0 \, \det(L_0 + X B_0 - T_0 E_0^{-1} T_0^T). \tag{5.10}$$

Here T_0, E_0 are rational, $E_0 > 0$, and B_0 is nonsingular. The primitive polynomial therefore has degree K, not the full matrix dimension. The rational factor $\det E_0$ disappears, whereas the rational correction inside the Schur complement remains. Consequently this reduction does not show that all benefit of the integer poles disappears.

This also explains a useful invariance. Adding an element of the full Z-radical to an extra lift is a determinant-one basis shear; it leaves the determinant polynomial exactly unchanged. More generally, rational changes of basis multiply it by a rational square and leave its primitive polynomial unchanged. Denominators introduced by a particular interpolation of the same subspace are not automatically an intrinsic determinant cost.

At the two checked indices $K = 5, 10$, the logarithms of the primitive values are respectively $101.17\ldots$ and $475.49\ldots$, compared with $94.29\ldots$ and $444.32\ldots$ for the radical alone. These finite comparisons motivate the lower-degree variant below; they do not establish an asymptotic failure theorem.

6 Divisibility of the low-degree determinants

The full radical in Section 5 forces degree K. We now omit it and retain only the added low-degree directions. Their number is $d = \lfloor K/5 \rfloor$. This changes both the determinant and its arithmetic; savings for the former full matrix cannot simply be transferred to this one.

6.1 The fixed lift and the statement

Use the J_m basis of Section 2. For the same block zero z from (5.6), define

$$U_{mi} = (-1)^{m-1} \frac{(K-a)!}{(K-m)!} \frac{(K+b)!}{(K+m)!} z_m, \quad a = 5i+1, \ b = a+4, \tag{6.1}$$

and set the other entries of column i to zero. The factorial ratios contain four factors, each at most $2K$. Thus

$$|U_{mi}| \le W_K = (2K)^4[15K(4K)^8]^2. \tag{6.2}$$

For $P_i = \sum_m U_{mi} J_m$, direct evaluation gives

$$m^2 P_i(-4m^2) = \tfrac{1}{2}\, 4^{K-1}(K-a)!(K+b)!z_m$$

on its block. The same isotropic equation therefore gives $U^T C_K U = 0$. The columns are independent and integral. Set

$$G_U(X) = U^T(R_K + XB_K)U, \quad D_U(X) = \det G_U(X), \quad F_U = q_U D_U, \quad A_U = q_U|\ell_U|,$$

where F_U is primitive and ℓ_U is the highest nonzero coefficient. The actual matrix is positive. We do not assume that $\deg D_U = d$.

Theorem 6.1. *Uniformly over the block choices satisfying (5.6)–(6.1),*

$$\limsup_{K\to\infty} \frac{\log A_U}{K^2} \le c_{\mathrm{ar}} = \frac{197 + 192\log 2}{150} = 2.2005617244\ldots. \tag{6.3}$$

The proof gives a finite coefficient-content divisor and an effective bound before taking this limit.

The argument has three distinct content gains. We describe them separately before combining them, so overlapping prime factors are not counted twice.

6.2 A global entry clearer, with the tails retained

The useful explicit odd residue is

$$\rho_{2j+1} = \frac{2(K-j)(K+j+1)}{4^{2K}} \binom{2K-2j}{K-j} \binom{2K+2j+2}{K+j+1}. \tag{6.4}$$

In particular it has no odd-prime denominator. Write $T_l = \rho_l V_{\cdot l} V_{\cdot l}^T$.

For $p > 2K$, collisions of the nodes $1, \ldots, 2K$ occur only in even–odd pairs with sum p, if they occur at all. Pairing their residues gives, over the localized rationals at p,

$$pM_K(X,Z) \equiv \left(4\chi X - \frac{7}{32}Z + \frac{1}{4}\sum_{v=1}^{(p-1)/2} \frac{(-1)^{v-1}}{v^4}\right) N_p \pmod p, \qquad \chi = (-1)^{(p-1)/2}. \tag{6.5}$$

To check the common scalar, put $j = (p-1)/2 - m$ in an even–odd pair. The rational part of the resolvent difference, after removing its common signed cubic factor, is

$$\tfrac{1}{4}\sum_{v=1}^{m} \frac{(-1)^{v-1}}{v^4} - 4\chi \sum_{v=0}^{j-1} \frac{(-1)^v}{(2v+1)^4}.$$

Adjacent values agree modulo p, since the two new denominators are opposite up to a factor two; at $m = (p-1)/2$ it is the sum in (6.5). The two rational endpoint corrections in the resolvents are what extend the even sum through m. All these tail denominators are below p. At paired roots, the two values p/D' are negatives modulo p, while the numerators agree. This proves (6.5). Its Z coefficient is a p-unit. Exact isotropy forces $U^T N_p U = 0$ modulo p, so every coefficient of G_U is p-integral. This conclusion applies after cancellation, not to arbitrary raw mixed minors.

For $K < p \le 2K$, set

$$Y_p = \sum_{p<l<2K,\ l \text{ odd}} b_l T_l, \qquad R_K = R_{0,p} - \chi p^{-4} Y_p.$$

The decomposition removes the only p-divisible denominator in each affected odd tail. The matrices $p^2 R_{0,p}$, $p^2 B_K$, $p^2 Y_p$ and $p^2 C_K/7$ are p-integral, and

$$p^2 Y_p \equiv 64\chi\, p^2 C_K/7 \pmod{p}. \tag{6.6}$$

Here division by seven is performed over $\mathbb{Q}$ before reduction, so the formula includes $p = 7$. For completeness, a possible upper odd node is $p+2v$. Its class contains $2v, p-2v, p+2v$, with a fourth node $2p-2v$ when $v \ge p-K$. Four-node classes have residue valuation at least -1. In a three-node class only the entry (v, v) can have valuation -2. The two local root-difference products, divided by p^2, are $-16v^2$ at $2v$ and $32v^2$ at $p+2v$. Their residue ratio is $-1/2$. Moreover $b_{p+2v} \equiv 4\chi(-1)^v v^3$ and $c_v/7 = (-1)^{v-1} v^3/32$, giving (6.6). All other residues disappear in this reduction. Compressing and using isotropy shows that $pY_{p,U}$, $p^5 R_U$ and $p^2 B_U$ are integral at p.

For the remaining sufficiently large odd primes, $p^2 > 4K$, six powers suffice even before compression:

$$p^6 R_K, \quad p^2 B_K, \quad p^2 C_K \quad \text{are } p\text{-integral.} \tag{6.7}$$

Every central binomial in (2.7) has valuation at most one, and every tail has valuation at least -4. If $p \nmid m$, this gives $v_p(e_m) \ge -2$, $v_p(r^e_m) \ge -4$. If $p \mid m$, the bounds are -4 and -1: the cubic numerator pays three tail powers and the endpoint costs one. For an odd node l with $p \nmid l$, each gap $(2m-l)(2m+l)$ costs at most one power, giving $v_p(T_l) \ge -2$. If $p \mid l$, two powers per gap are possible, but $v_p(b_l) = 3$ and $v_p(r^o_l) \ge -1$. These four cases prove (6.7), including the endpoint terms.

Put $k_p = \lfloor \log_p(4K) \rfloor$ and

$$Q_K = 2^{4K+4+8k_2} \prod_{\substack{3\le p\le K\\ p^2\le 4K}} p^{8k_p} \prod_{\substack{3\le p\le K\\ p^2>4K}} p^6 \prod_{K<p\le 2K} p^5, \tag{6.8}$$

with products over primes. Then $Q_K R_U, Q_K B_U$ are integral. For small odd primes, each of the two binomial valuations and each linear gap costs at most k_p; the tails cost at most $4k_p$. Thus $8k_p$ suffices. At two the odd gaps are units, the denominator of ρ_l costs at most $4K-1$, and the finite tails and even terms are covered by $4 + 8k_2$. This proves the remaining factors. The prime number theorem gives

$$\log Q_K = (11 + 4\log 2 + o(1))K. \tag{6.9}$$

The small primes cost $O(\sqrt{K}\log K)$; the two large-prime ranges have weights six and five.

6.3 Exact tail rank gives three powers per missing direction

For $K < p \le 2K$, the number of terms in Y_p is $t_p = K - (p+1)/2$. They are rank one over $\mathbb{Q}$. The exact identity

$$p^5 G_U(X) = -\chi A_p + p^3 S_p(X), \qquad A_p = pY_{p,U}, \quad S_p = p^2(R_{0,p,U} + XB_U)$$

has p-integral entries and $\operatorname{rank} A_p \le t_p$ over $\mathbb{Q}$. In the column expansion of the determinant, more than t_p columns from A_p give zero identically. Every surviving term therefore contains at least $(d-t_p)_+$ columns from $p^3 S_p$. It follows that

$$\mathcal{T}_K = \prod_{K<p\le 2K} p^{3(d-t_p)_+} \ \text{ divides } \ \operatorname{cont}(Q_K^d D_U), \qquad \log \mathcal{T}_K = (3/25 + o(1))K^2. \tag{6.10}$$

The rate is $3\int_{8/5}^{2}(x/2-4/5)\,\mathrm{d}x = 3/25$. Rank only modulo p would not justify three powers per direction; the exact rational rank is what makes that conclusion possible.

6.4 A common dyadic factor

Set $\eta_0 = \eta_1 = 0$ and

$$\eta_s = \sum_{j=1}^{s-1}(8j - 1 - 2s_2(j)), \qquad s \geq 2. \tag{6.11}$$

Every s-minor of $V_{ml} = (4m^2 - l^2)^{-1}$ has dyadic valuation at least η_s. The Cauchy determinant denominator is odd. For its numerator use, for $j \geq 1$, the integer-valued polynomials

$$\frac{\prod_{i=0}^{j-1}(t - i^2)}{j(2j-1)!}, \qquad \frac{\prod_{i=0}^{j-1}(t - (2i+1)^2)}{4^j(2j)!}.$$

Take the degree-zero member of each sequence to be 1. At $t = m^2$ and $t = (2k+1)^2$ their values are respectively $\binom{m+j}{2j} + \binom{m+j-1}{2j}$ and $\binom{k+j}{2j}$. Evaluation determinants therefore divide the even and odd Vandermondes by the factors $\prod_{j=1}^{s-1} 4^j j(2j-1)!$ and $\prod_{j=1}^{s-1} 4^j(2j)!$. Their total dyadic valuation is (6.11).

Cauchy–Binet preserves this bound in $U^T V$. Write the pencil scaled by $2^{4K+4+8k_2}$ as the sum of its odd-residue Gram part and its even rational part. The first has integral diagonal weights, and every s-minor is divisible coefficientwise by $2^{2\eta_s}$. The formulas (2.7), (2.4), and $v_2\binom{2n}{n} = s_2(n)$ show that every entry of the even part has valuation at least $8K - 2$. Since $d \leq K/5$,

$$2(\eta_d - \eta_s) \leq 16(d-1)(d-s) \leq (8K-2)(d-s).$$

The complementary-minor expansion consequently proves

$$2^{2\eta_d} \mid \mathrm{cont}(Q_K^d D_U), \qquad 2\eta_d \log 2 = (8\log 2/25 + o(1))K^2. \tag{6.12}$$

This applies to every coefficient, not just the leading one.

6.5 Two valuation layers below the pole cutoff

Fix an odd prime $p \leq K$ with $p^2 > 4K$. In a nonzero square class of nodes $l^2 \bmod p$, let e, o count even and odd nodes, and let ϵ_m indicate whether $2m$ belongs to the class. Every root difference within that class has valuation exactly one. Counting factors in $J_m J_n / D'$ gives the residue valuations

$$e - o + 1 - \epsilon_m - \epsilon_n \quad \text{at odd nodes}, \qquad e - o - 1 \quad \text{at an even diagonal node}. \tag{6.13}$$

Write $n = \lfloor 2K/p \rfloor$, $s = 2K - np$ and $a = \min(s, p - s - 1)$. Represent a nonzero class by $1 \leq r < p/2$. Its $2n$ base nodes $r + jp, p - r + jp$ $(0 \leq j < n)$ are balanced in parity. Exactly one extra node occurs precisely for $1 \leq r \leq a$; its parity is that of $n + r$. Let I_p be the even cofactor indices in odd-majority classes and Z_p those in even-majority classes. Zero-class indices belong to neither set. If N_I, N_Z count the two parities of r, then

$$|I_p| = nN_I, \quad |Z_p| = (n+1)N_Z, \quad N_I + N_Z = a, \quad |N_I - N_Z| \leq 1.$$

The sets are unions of at most n and $n+1$ integer intervals. For example, in $2K/3 < p \leq K$ a class first has four balanced nodes; a fifth makes it unbalanced and a sixth restores balance. Only odd-majority classes contribute the worst rational layer.

Finite tails cost at most four powers. Formula (6.13) therefore gives $v_p(R_{mn}) \geq -6$ everywhere, at least -5 if one index is outside I_p, and at least -4 if one index is in Z_p and the other is outside I_p. The zero class needs a separate check: it consists of $p, 2p, \ldots, np$, with squared differences of valuation two. Double the counts in (6.13); the cubic resolvent factor pays three tail powers. All its rational contributions are at least -5, and those involving a Z_p index are at least -3. Thus they respect the stated bounds. The beta matrix has valuation at least -2 and does not weaken them.

Let b_p count fixed five-node blocks meeting I_p, let z_p count blocks wholly in Z_p, and put $c_p = d - b_p - z_p$. After compression, the coefficientwise lower valuation table of $p^6 G_U$ is

	I	remaining	Z
I	0	1	1
remaining	1	1	2
Z	1	2	2

.

Divide every Z row and column by p, and then every remaining row by p. The result is still integral at p. Hence

$$\Gamma_K = \prod_{\substack{p \leq K,\ p \text{ odd} \\ p^2 > 4K}} p^{d - b_p + z_p} \quad \text{divides } \operatorname{cont}(Q_K^d D_U). \tag{6.14}$$

This exponent already includes the first-layer rank defect $d - b_p$; that factor must not be multiplied in again.

Interval counting gives, uniformly,

$$d - b_p + z_p = K/5 + a/10 + O(K/p).$$

The weighted error sums to $o(K^2)$. For $x = p/K$, the continuous version of a/K is $f(x) = \min(2 - nx, (n+1)x - 2)$ with $n = \lfloor 2/x \rfloor$. Its graph on $2/(n+1) < x \leq 2/n$ is a triangle. Therefore

$$\begin{aligned} \lim K^{-2} \log \Gamma_K &= \frac{1}{5} + \frac{1}{10} \int_0^1 f(x)\, \mathrm{d}x \\ &= \frac{1}{5} + \frac{1}{10} \sum_{n=2}^{\infty} \frac{2}{n(n+1)(2n+1)} = \frac{23}{30} - \frac{4}{5} \log 2. \end{aligned} \tag{6.15}$$

Partial summation uses the prime number theorem on $\varepsilon K < p \leq K$; the omitted content is at most $2d\vartheta(\varepsilon K)$, so $\varepsilon \downarrow 0$ justifies the endpoint. The series follows by partial fractions and harmonic number differences.

6.6 Primitive arithmetic and the compatible comparison

The three content factors (6.10), (6.12), and (6.14) have disjoint prime supports. To bound coefficient sizes, use (6.4) and $\binom{2n}{n} \leq 4^n$ to get $\rho_l \leq 8K(K+1) \leq 16K^2$. The alternating finite tails have absolute value at most one, so $|b_l| \leq 4K^3$ and $|r_l^o| \leq 5K^3$. Since every gap has magnitude at least one, the total odd contribution is at most $64K^6$ in B_K and $80K^6$ in R_K. The reverse binomial bound gives

$$|e_m| \leq \frac{(2K+1)^2}{8m^2}, \qquad |r_m^e| \leq \frac{3m^3}{8},$$

so the even contribution is at most K^3. Therefore $|R_{mn}|, |B_{mn}| \leq 81K^6$, and every compressed entry has magnitude at most $M_K = 2025 W_K^2 K^6$, and every determinant coefficient is bounded by $2^d d! M_K^d$. It follows that

$$A_U \leq \frac{Q_K^d\, 2^d d! M_K^d}{\mathcal{T}_K\, 2^{2\eta_d} \Gamma_K}. \tag{6.16}$$

This includes primes from the numerator of ℓ_U, regardless of its degree. Combining the rates proves Theorem 6.1.

To interpret it, put $\delta_U = D_U(\beta(4))/|\ell_U|$. An irrationality argument would follow from $A_U\delta_U \le e^{-\varepsilon K^2}$ along an unbounded sequence. The available analytic restriction is

$$\limsup \frac{-\log \delta_U}{K^2} \le \frac{\pi/2 + 2\log(5/4) + 2\arctan 2 + 4\log(2+\sqrt{5})}{5} = 2.0011845531\ldots. \tag{6.17}$$

Appendix B proves it for every degree-below-K subspace of dimension at most $K/5$, without a beta-rank assumption. The arithmetic upper bound $2.20056\ldots$ is above this ceiling. Thus these particular estimates cannot establish the desired comparison. They do not supply a necessary arithmetic cost, so they do not prove that the primitive determinants themselves fail.

7 Thirteen beta values and a rational Gram block

This construction has a different advantage: its polynomial degree stays bounded while its matrix size grows. It involves precisely $\beta(4), \beta(6), \ldots, \beta(28)$. The question is whether the growing matrix supplies useful decay after its rational part has been removed. The one-label antecedent is the ordinary-moment Schur construction in [8, Section 8]; we add seven logarithmic labels.

7.1 The seven boundary functions

On $0 < x < 1$, put $L = -\log x$, $t = x^2$, and $\mathrm{d}\mu = L^3\,\mathrm{d}x/(1+x^2)$. For $r \ge 1$, take the Gram matrix of

$$t^j L^{2a}, \qquad 0 \le a \le 6, \quad 0 \le j < r.$$

These functions are independent, as is seen by substituting $x = e^{-y}$ and using independence of exponential polynomials. The matrix is positive definite. For $s = 4 + 2a + 2b$, its entries are

$$(s-1)!(-1)^{j+k}\left(\beta(s) - \sum_{v=0}^{j+k-1} \frac{(-1)^v}{(2v+1)^s}\right).$$

The formula follows by a finite geometric identity and integration of $x^m(-\log x)^{s-1}$.

For each logarithmic label replace $1, t, \ldots, t^{r-1}$ by $1, (1+t), (1+t)t, \ldots, (1+t)t^{r-2}$. This change has determinant one. Grouping the seven boundary functions L^{2a} first gives

$$\begin{pmatrix} A & C \\ C^T & B \end{pmatrix}, \qquad S_r = A - CB^{-1}C^T, \qquad \det G_r = \det B \det S_r. \tag{7.1}$$

Write X_k for $\beta(4+2k)$, $0 \le k \le 12$. The blocks are

$$\begin{aligned} A_{ab} &= (3+2a+2b)!X_{a+b}, \\ C_{a,(b,j)} &= \frac{(3+2a+2b)!}{(2j+1)^{4+2a+2b}}, \\ B_{(a,i),(b,j)} &= (3+2a+2b)!\left((2i+2j+1)^{-4-2a-2b} + (2i+2j+3)^{-4-2a-2b}\right), \end{aligned}$$

where $0 \le i, j \le r-2$. For $r = 1$ the bulk is empty. B is rational positive definite, so the primitive polynomial associated with $\det G_r$ is that associated with $P_r(X) = \det S_r(X)$. The rational factor $\det B$ supplies no separate gain. This exact reduction alone does not determine the residual decay.

The one-label case at $r = 2$ makes the normalization visible: $A = 6\beta(4)$, $C = 6$, and $B = 164/27$, so

$$S = 6\beta(4) - \frac{243}{41} = \frac{3}{41}(82\beta(4) - 81).$$

The primitive polynomial is $82X - 81$. The factor from the rational bulk cannot make its primitive value smaller.

7.2 The residual decays on a linear scale

Let $\rho = 3 + \sqrt{8}$ and let T_j denote the Chebyshev polynomial. Define

$$H_0 = \prod_{a=0}^{6}(3+4a)!, \quad V_0 = \pi^{84}\left(\prod_{j=1}^{6}(2j)!\right)^2, \quad M_0 = \sqrt{1+(13\pi/2)^2}, \quad \Gamma = 28 + 28\sqrt{26M_0} < 700.$$

Theorem 7.1. *At the actual beta values, for every $r \geq 1$,*

$$\frac{V_0 e^{-\Gamma r}}{(52r^2)^7} \leq P_r(\beta) \leq \frac{H_0}{T_{r-1}(3)^{14}} \leq 2^{14} H_0 \rho^{-14(r-1)}. \tag{7.2}$$

In particular $\log P_r(\beta) = -\Theta(r)$ *and* $r^{-2}\log P_r(\beta) \longrightarrow 0$.

Proof. The Schur complement expresses projection onto the orthogonal complement of the bulk. For any real $v = (v_a)$,

$$v^T S_r v = \min_{p_a(-1)=v_a,\ \deg p_a < r} \int_0^1 \left|\sum_{a=0}^{6} p_a(x^2)L^{2a}\right|^2 \mathrm{d}\mu.$$

The constraint says exactly that $p_a(t) = v_a - (1+t)Q_a(t)$ for a bulk polynomial Q_a. Choose $p_a(t) = v_a T_{r-1}(2t-1)/T_{r-1}(-3)$. Its common factor is at most $1/T_{r-1}(3)$ on $[0,1]$. Hence $0 < S_r \leq A/T_{r-1}(3)^2$ in matrix order. Take determinants, apply Hadamard to A, and use $\beta(s) < 1$ to obtain the upper bound. The formula $T_j(3) = (\rho^j + \rho^{-j})/2$ gives the final inequality.

For the lower bound, write $y = -\log x$. Every residual lies in

$$\mathcal{E}_r = \mathrm{span}\{y^h e^{-2jy} : 0 \leq j < r,\ 0 \leq h \leq 12\}, \quad \|F\|_\mu^2 = \int_0^\infty |F(y)|^2 \frac{y^3 e^{-y}}{1+e^{-2y}}\,\mathrm{d}y.$$

At the seven points $z_l = i\pi(l+1/2)$, the bulk factor $1 + e^{-2z_l}$ vanishes. The residual R_a of the boundary function y^{2a} therefore satisfies $R_a(z_l) = z_l^{2a}$, and

$$|\det(R_a(z_l))_{l,a=0}^{6}|^2 = V_0. \tag{7.3}$$

This is a Vandermonde identity: $\prod_{i<j}(j-i)(j+i+1) = \prod_{j=1}^{6}(2j)!$.

The repeated-exponent evaluation estimate proved in Appendix C gives, for $F \in \mathcal{E}_r$,

$$|F(z_l)|^2 \leq 52r^2 e^{\Gamma r/7}\|F\|_\mu^2.$$

Orthonormalize the seven residuals. Each row of their evaluation matrix has squared length at most this evaluation-norm bound. Hadamard bounds its squared determinant by the seventh power. Undoing the basis change and using (7.3) gives the lower bound in (7.2). □

7.3 The arithmetic question left by this reduction

Let $q_r > 0$ make $q_r P_r$ primitive integral. Its total degree is seven. The coefficient of X_6^7 is $-(15!)^7$: only the reversing permutation $a \mapsto 6-a$ contributes. Thus $q_r \geq 1/(15!)^7$. Under joint rationality of the thirteen values, a fixed common denominator raised to the seventh power would make every positive $q_r P_r(\beta)$ an integer. It is therefore sufficient to prove $q_r P_r(\beta) \to 0$ on an unbounded subsequence.

There are useful exact bounds in both directions. Let $\ell(m)$ be the lcm of the positive odd integers at most m. For $r \geq 2$, put

$$H_{aa} = \ell(2r-3)^{2+2a}, \qquad E_{(a,i),(a,i)} = \ell(2i+2r-1)^{2+2a}.$$

The matrices $M = EBE$, $N = HCE$ and $HA(X)H$ are integral: split each denominator exponent $4+2a+2b$ between its two indices. The block determinant

$$\det \begin{pmatrix} HAH & N \\ N^T & M \end{pmatrix} = U_r P_r(X), \qquad U_r = (\det H)^2 \det M \in \mathbb{Z}_{>0}$$

therefore gives $q_r \leq U_r$. Hadamard and $\sum_{a=0}^{6}(2+2a) = 56$ imply

$$U_r \leq C_0^{r-1} \left(\ell(2r-3) \prod_{i=0}^{r-2} \ell(2i+2r-1) \right)^{112}, \quad C_0 = \prod_{a=0}^{6} 2(3+4a)!.$$

The prime number theorem gives $\log q_r \leq 336r^2 + o(r^2)$. This upper bound is too large for the linear-scale sufficient condition, but says nothing about whether the true cost is quadratic.

For a necessary condition, put $R_r = CB^{-1}C^T$ and $F_7 = \prod_{a=1}^{6}(3+4a)!$. The degree-six coefficient

$$[X_2X_4X_6X_8X_{10}X_{12}]P_r = -F_7(R_r)_{00} \tag{7.4}$$

isolates one rational entry. Give X_k weight k. The cofactor of entry (i,j) of A has weight $42-i-j$; only $(0,0)$ can contribute to the displayed weight 42. In that cofactor the successively smallest variables force the diagonal, proving (7.4). If b_r is the reduced denominator of $(R_r)_{00}$ and $c_7 = (15!)^7$, then integrality of both coefficients gives

$$q_r \geq \frac{b_r}{c_7 \gcd(b_r, F_7)}.$$

A superlinear lower bound for $\log b_r$ would rule out successful shrinking on that sequence. None has been proved here.

Along any chosen unbounded sequence of indices, Theorem 7.1 gives the sufficient condition $\limsup \log q_r/r < 14 \log \rho = 24.67846\ldots$, whereas convergence of the primitive values to zero requires $\limsup \log q_r/r \leq \Gamma$. Neither arithmetic bound decides this comparison.

r	$\log q_r$	$\log P_r(\beta)$	$\log(q_r P_r(\beta))$
1	−152.01	194.93	42.92
2	14.04	−3.33	10.71
3	655.17	−119.25	535.92
4	1586.31	−254.99	1331.32
5	3103.65	−402.44	2701.21
6	4842.52	−545.27	4297.25

These certified finite values all exceed one after primitive normalization. They illustrate the arithmetic issue, but do not prove growth along every unbounded subsequence. The general conclusion is the residual scale theorem, with primitive arithmetic unresolved.

8 Complete cancellation collections for fourteen values

We now return to linear forms. The target set is $\{\beta(4), \beta(6), \ldots, \beta(30)\}$, with fourteen members. Fifteen independent forms on these constants together with 1 cannot all vanish. This collective nonvanishing route avoids having to identify one particular form that remains nonzero. Its unresolved cost is the height of the entire collection.

The rational-function construction adapts Krisman's hypergeometric method [5] and the isolated-pole refinement in [10]. We use twenty-nine nested pole blocks, double integer zeros, and one derivative. These choices differ from the thirty-one blocks, triple zeros, and two derivatives in the fifteen-value result.

8.1 Formal rank from a rational antidifference

It is convenient to prove rank for general nested blocks. Let $n > 0$ be even, let b be a nonnegative integer, and let S be a positive odd integer. Choose integers $0 \le d_j \le n/2$ for $1 \le j \le S$. Put $L = n + 2b$ and

$$N(t) = \prod_{\ell=-n-b-1}^{b} (t-\ell)^2, \quad Q(t) = \prod_{j=1}^{S} \prod_{a=d_j}^{n-d_j} (t + a + \tfrac{1}{2}), \quad R(t) = N(t)/Q(t).$$

For a polynomial invariant under $t \mapsto -n-1-t$, with PR proper, define

$$\mathcal{J}(P) = \sum_{t \ge 0} (-1)^t (PR)'(t).$$

The differentiated series converges absolutely. Partial fractions give rational formal coefficients on $1, \beta(2), \ldots, \beta(S+1)$. A *formal zero* means that all these coefficients vanish, without making any assumption about relations among the actual constants.

Theorem 8.1. *If* $\deg P \le S+1$*, a formal zero forces* $P = 0$*. More generally, subject to properness, all formal-zero symmetric polynomials are*

$$P(t) = g(t)B(t-1) + g(t+1)A(t), \qquad g(-n-t) = -g(t), \tag{8.1}$$

where

$$A(t) = (t+n+b+2)^2 \prod_j (t + d_j + \tfrac{1}{2}), \quad B(t) = (t-b)^2 \prod_j (t + n - d_j + \tfrac{3}{2}).$$

The first possible nonzero formal identity has degree $S+3$*.*

Proof. Write $F = PR = \sum_{a,i} c_{i,a}(t + a + 1/2)^{-i}$, with pole orders $B_a = \#\{j : d_j \le a \le n - d_j\}$. Reflection gives $c_{i,n-a} = (-1)^{i+1} c_{i,a}$. The coefficient of $\beta(i+1)$ is $-i2^{i+1} \sum_a (-1)^a c_{i,a}$. Reflection already makes this sum zero for even i. If the other beta coefficients vanish too, all alternating residue sums vanish.

Define $h_{i,-1} = 0$, $h_{i,a} = c_{i,a} - h_{i,a-1}$. Then $h_{i,n} = 0$, and matching principal parts gives the unique proper rational function

$$H(t) = \sum_{a=0}^{n-1} \sum_i \frac{h_{i,a}}{(t + a + 1/2)^i}, \qquad F(t) = H(t) + H(t+1).$$

Uniqueness follows because a nonzero rational solution of $H(t) + H(t+1) = 0$ would have infinitely many translated poles, or would be a polynomial whose leading terms cannot cancel.

The forward and backward recurrences bound the pole order at $-a-1/2$ by $\min(B_a, B_{a+1})$, since the orders are unimodal. Thus

$$H = V/Q_H, \quad Q_H = \prod_j \prod_{a=d_j}^{n-d_j-1} (t+a+1/2), \quad \deg Q_H = \deg Q - S.$$

This loss of one factor per block is the rank mechanism. A nonzero H has the same degree at infinity as F, because its leading term doubles when shifted and added. Consequently $\deg V = 2L+4+\deg P-S$.

Uniqueness and reflection imply $H(-n-t) = -H(t)$, so $H(-n/2) = 0$. The double zeros of F imply the two recurrences $H(\ell) + H(\ell+1) = 0$ and $H'(\ell) + H'(\ell+1) = 0$ on the integer-zero interval. The first propagates from the center to every integer $-n-b-1, \dots, b+1$. The second needs the rational constant: telescoping gives $\mathcal{J}(P) = H'(0)$, so a formal zero gives $H'(0) = 0$ and the same propagation for derivatives. Thus V has $L+3$ double zeros, total degree at least $2L+6$. If $\deg P \le S+1$, its degree is at most $2L+5$, a contradiction unless $P = 0$.

In general those double zeros give $H = g(t)(t-b-1)^2 N(t)/Q_H(t) = g(t)B(t-1)R(t)$. Reflection makes g odd about $-n/2$, and $R(t+1)/R(t) = A(t)/B(t)$ gives (8.1). Conversely that formula yields $PR = H + H(t+1)$; properness makes H proper, the alternating residue sums vanish, and $H'(0) = 0$ because of its double zero there. Hence every formal coefficient vanishes. Finally the leading terms in (8.1) add, giving $\deg P = \deg g + S + 2 \ge S + 3$. □

8.2 The sixteen coefficient directions and integral reduction

Specialize to $116 \mid n$, $b = n$, $S = 29$, and offsets $d_j = jn/116$ for $0 \le j \le 28$. Then $\deg N = 6n+4$ and $\deg Q = 22n+29$. Put $x = 2t+n+1$. The sixteen multipliers $1, x^2, \dots, x^{30}$ give an invertible rational coefficient matrix on $1, \beta(2), \dots, \beta(30)$ by Theorem 8.1. The beta-two cancellation space has dimension fifteen, and every independent collection of fifteen cancellations gives an invertible matrix on $1, \beta(4), \dots, \beta(30)$.

Higher degrees add known identities rather than new coefficient directions. Write $X_n = 3n+3$ and let $E_h(x)$ be the even part of

$$(x-1)^{2h+1}(x-X_n)^2 \prod_{j=0}^{28}(x+n-2d_j), \qquad h \ge 0.$$

Taking an even part just discards odd coefficients. Thus E_h is monic in $\mathbb{Z}[x^2]$, of degree $h+16$ in x^2. Formula (8.1) shows that it is a formal zero whenever the product is proper. Subtracting integer multiples of these monic identities reduces every integer even polynomial of degree $2D \le 2n$ uniquely to degree at most thirty, with the same coefficient vector. This is an additive quotient; multiplying an arbitrary identity by x^2 need not preserve it.

For the weighted coefficient norm $\|P\|_{X_n} = \sum_k |[x^k]P| X_n^k$,

$$\|E_h\|_{X_n}/X_n^{2h+32} \le 4(1+1/X_n)^{2h+1}(1+n/X_n)^{29} < 40000$$

in this range. Successive monic elimination therefore gives $\|P_{\mathrm{red}}\|_{X_n} \le 40001^{\max(0,D-15)}\|P\|_{X_n}$. For $D = o(n)$ this costs only $\exp(o(n))$, and introduces no denominators. It does not ensure that short vectors have independent nonzero reductions.

8.3 Eleven directions with uniform height and decay

For $P(x) = \sum_{j=0}^{15} w_j x^{2j}$ put $h_n(P) = \sum_j |w_j|(3n+3)^{2j}$, and let $\Lambda_n \subset \mathbb{Z}^{16}$ be the lattice canceling the beta-two coefficient. Its rank is fifteen. The following two quantitative estimates use different clearing integers: one bounds the primitive normal to this lattice, the other clears the actual forms.

Lemma 8.2. *Let a_n be the primitive integer normal to Λ_n. There is a constant $H_* < 24.522$ such that*

$$\log \|a_n\|_\infty \le (H_* + o(1))n.$$

There are common positive integers D_n^* clearing every coefficient of every allowed beta-two-canceling form, for which

$$|D_n^* \mathcal{J}_n(P)| \le h_n(P) \exp((-C_* + o(1))n), \qquad C_* > 6.0384.$$

The error terms are uniform in the multiplier coefficients.

Appendix D gives the arithmetic and analytic estimates, including the finite rational certificate defining these constants. The parity grouping used for a_n is especially useful: the even-address and odd-address denominator products are coprime modulo two, so their grouped residue sums have much better dyadic integrality than individual residues.

Theorem 8.3. *For every sufficiently large positive multiple n of 116, there is an integer matrix C_n with eleven rows and fifteen columns such that, for $\theta = (1, \beta(4), \beta(6), \dots, \beta(30))^T$,*

$$\operatorname{rank}_{\mathbb{Q}} C_n = 11, \qquad \|C_n\theta\|_\infty \le e^{-1.12n}.$$

Its rows come from eleven independent beta-two-canceling multipliers of weighted height at most $e^{4.91n}$.

Proof. The lattice with primitive normal a_n has Euclidean covolume $\|a_n\|_2$. After applying $T = \operatorname{diag}(1, X_n^2, \dots, X_n^{30})$, its covolume is

$$\det T \, \|T^{-1}a_n\|_2 \le 4X_n^{240}\|a_n\|_\infty.$$

The classical LLL product estimate [6, Proposition 1.6] gives a basis whose product of lengths is at most $2^{15\cdot 14/4}$ times this covolume. To recall the relevant estimate, size reduction gives $|\mu_{ij}| \le 1/2$ and the reduction condition gives $\|b_i^*\|^2 \ge \|b_{i-1}^*\|^2/2$. Expanding the basis vectors in their orthogonal components and multiplying yields the stated product bound.

Order its lengths as $s_1 \le \cdots \le s_{15}$. Each is at least one since the weighted lattice is integral, and $s_k^{16-k} \le \prod_i s_i$. Weighted ℓ^1 height is at most four times Euclidean length. Thus the kth weighted successive minimum satisfies

$$\limsup n^{-1} \log \lambda_k(n) \le H_*/(16-k).$$

At $k = 11$ this is below 4.905, so eleven independent vectors have height at most $e^{4.91n}$ for all sufficiently large admissible n. Apply the *same* D_n^* from Lemma 8.2 to their forms. Since $6.0384 - 4.91 > 1.12$, their values satisfy the claimed bound. The invertible formal coefficient map preserves their independence after cancellation and common clearing. □

An eleven-by-fifteen matrix can annihilate θ exactly, so Theorem 8.3 does not establish irrationality. Fifteen independent small rows would exclude that possibility. The product argument above gives only the twelfth-minimum upper rate $H_*/4 = 6.13044\ldots$, already above C_*. This is the limitation of this height estimate, not a lower bound on the true twelfth or fifteenth minimum.

At $n = 116, 232, 464$, exact finite computations do give complete fifteen-row collections, with largest primitive values below 10^{-236}, 3.68×10^{-470}, and 6.04×10^{-890}, respectively. These are the same kind of objects needed for the contradiction. The unproved step is their uniform height or decay along an unbounded sequence. A bound for one shortest vector is insufficient, and the available covolume bound does not give the required rate: the kernel of $(1, M, 0, \dots, 0)$ has many unit directions while its last direction must still have size comparable to M.

9 Other tests and the remaining questions

9.1 A recurrence can preserve too much

The finite complete bases in Section 8 suggested transporting them between neighboring indices. There is an elementary test that such a proposal must pass before estimating transition matrices. Suppose fixed rational rows $v_1, \dots, v_r$ span $(1, 0, \dots, 0)$, and every nonzero integer row $c_{i,n}$ is proportional to its corresponding v_i. Replace each v_i by its primitive integer representative u_i. Then $c_{i,n} = k_{i,n} u_i$ with nonzero integral $k_{i,n}$. For $\theta = (1, \theta_1, \dots)$, at least one fixed $u_i\theta$ is nonzero, and

$$|c_{i,n}\theta| \geq |u_i\theta| > 0.$$

Such transport cannot make all the values tend to zero. The proposed neighboring-index update preserved precisely these individual coefficient directions. A recurrence that supplies new combinations or an additional quantitative constraint is not excluded; merely propagating the same rational lines cannot solve the height problem.

9.2 A Loewner specialization with an inconclusive cost comparison

We also considered the divided-difference construction of Litsey [7], specialized to

$$F(z) = \frac{\psi''(z + 1/4) - \psi''(z + 3/4)}{1536}, \qquad F'(0) = \beta(4).$$

Here $\psi = \Gamma'/\Gamma$ is the digamma function. The two cubic pole classes of F give a degree-six difference equation. The specialization of the cited nonvanishing argument applies for $r \geq 4$, $m \geq 0$ and every rational evaluation point: some minor deleting at most three common nodes is nonzero. The opposite signs of the two pole classes do not invalidate that argument.

One fixed geometry, $r = 200k$, $m = 13k$, was assessed on the vertical contour $\Re z = -1/8$, between the nonnegative nodes and the negative poles. It avoids enclosed-pole terms. The moving-contour and deletion errors are uniform lower-order terms, including every position of the at most three deletions. On the r^2 scale the certified lower decay rate was greater than 2.488, while the selected common clearer had upper rate 19.2552. These bounds establish no useful comparison. Neither is asserted to be optimal or necessary; the outcome is inconclusive. This assessment is retained in the record catalog rather than promoted to a failure theorem for the Loewner construction.

9.3 What would change the conclusions

The results leave three different kinds of question. For the linear forms, uniform formal rank is established and eleven directions have sufficient height control; the missing condition is a complete useful collection, or nonvanishing for the same small forms. For the thirteen-value Gram construction, the residual analytic scale is known, and the missing input is its primitive arithmetic. For the low-degree mixed-pole determinants, genuine uniform content savings are known, but the true arithmetic and achieved decay have not been compared sharply enough.

The final substitution result is different. Its necessary dyadic cost and the uniform lower bound for its error envelope prove that the specified certificate fails. Improving a convenient entry clearer cannot evade that theorem. A successful modification would have to change an assumption of the certificate, such as use of the global polynomial clearer or the generic q^{-2} error envelope, and then establish the corresponding new arithmetic and approximation estimates.

These distinctions explain the value of retaining both successful lemmas and unsuccessful constructions. They identify which parts can be reused and which estimates would actually move the argument forward.

9.4 Prepared with Codex

Prepared with OpenAI Codex, using GPT-6-Astra at maximum reasoning effort. The cooperating sessions developed and cross-checked the underlying records; this manuscript assembles their mathematical results.

9.5 Verification

The manuscript passed a separate cooperating-session mathematical and exposition review. The accompanying records state the review and finite checks' scope; neither constitutes formal verification.

A Larger fixed isotropic blocks

Fix an odd $L \geq 5$ before letting K grow. In a block of length L, (5.5) contains $2L-2$ factors, so the quadratic coefficients are bounded by $H = K(4K)^{2L-2}$. The alternating diagonal form has balanced signature. We need a bound for a whole isotropic collection, rather than repeated applications of a bound for one vector without tracking the intervening basis changes.

For a nonsingular integral symmetric N-by-N matrix of balanced signature, odd $N \geq 5$, with entries bounded by $H \geq 1$, set

$$V_N = (3N^2H)^{(N-1)/2}, \quad F_N = 8N^2H^2V_N^2, \quad H'_N = N^2HF_N^2.$$

Define $T_5(H) = V_5$ and $T_N(H) = \max(V_N, NF_NT_{N-2}(H'_N))$. For fixed N this is polynomial in H. We claim that $(N-3)/2$ independent mutually orthogonal integral isotropic vectors can be chosen with sup norm at most $T_N(H)$.

For $N = 5$ this is Meyer and Cassels. In higher dimension choose a nonzero isotropic v with norm at most V_N, and a coordinate j such that $b = v^TAe_j \neq 0$. Put $w = 2be_j - A_{jj}v$. Then $w^TAw = 0$ and $v^TAw = 2b^2$, so v, w split a rational hyperbolic plane. The integral vectors

$$f_i = 2b^2e_i - (e_i^TAw)v - (e_i^TAv)w$$

span its orthogonal complement and have coordinates bounded by $(6N^2+2N)H^2V_N^2 \leq F_N$. Select $N-2$ independent columns to form F. The induced matrix F^TAF is nonsingular, balanced, and bounded by H'_N. Apply induction there and lift back by F, retaining v as the first isotropic vector. This proves the claimed recursion and controls the entire collection.

Applying this construction in the fixed L-blocks and then using (5.7) gives integral cofactor coordinates of size $K^{O_L(1)}$. Hadamard bounds $\det(\mathcal{U}^T\mathcal{U})$ even when supports overlap within a block. Cauchy–Binet and $\sum_I |\det \mathcal{U}_I| \leq \sqrt{\binom{2K}{h}\det(\mathcal{U}^T\mathcal{U})}$ give $O_L(K\log K)$ logarithmic transfer cost. The resulting dimension is $K + (L-3)\lfloor K/L \rfloor/2$. No uniformity for growing L is asserted.

B The normalized-decay ceiling

Let P be any real polynomial of degree below K, with

$$G(P) = \int_0^\infty \frac{P(4y^2)^2 w(y)}{D(4y^2)}\,\mathrm{d}y, \qquad B(P,P) = \sum_{l \text{ odd}} \frac{b_l P(-l^2)^2}{D'(-l^2)}.$$

The weight bound in Lemma 4.1 gives $w(y) \ge c_0 e^{-\pi y}$ with $c_0 = 1/(48\pi)$. Put $p(t) = P(K^2 t)$. Restriction to $0 \le y \le K/2$ yields

$$G(P) \ge \frac{c_0 K e^{-\pi K/2}}{2D(K^2)} \int_0^1 p(x^2)^2 \, \mathrm{d}x.$$

Let P_j be the standard Legendre polynomial, normalized by $\mathsf{P}_j(1) = 1$. The shifted functions $\phi_j(t) = \sqrt{2j+1}\mathsf{P}_j(2t-1)$ are orthonormal for Lebesgue measure on $[0,1]$. The classical integral formula at $x > 1$ [3, 14.12.7], together with parity and continuity, gives $|\mathsf{P}_j(-1-2a^2)| \le (9+\sqrt{80})^j$ for $0 \le a \le 2$. Also $\int_0^1 p(t)^2 \, \mathrm{d}t \le 2\int_0^1 p(x^2)^2 \, \mathrm{d}x$. Since $\sum_{j=0}^{K-1}(2j+1) = K^2$, Cauchy–Schwarz in this orthonormal basis gives

$$|P(-l^2)|^2 \le \frac{4K}{c_0} e^{\pi K/2} D(K^2)(2+\sqrt{5})^{4K-4} G(P).$$

Let $N = 2K$. Formula (5.2) implies $\min_l |D'(-l^2)| \ge (N!)^2/(2N^2)$. The increasing function $\log(1/4 + x^2)$ has right-sum error at most $\log 5$. Since its integral on $[0,1]$ is $\log(5/4) - 2 + \arctan 2$ and $N! \ge (N/e)^N$,

$$\frac{D(K^2)}{\min_l |D'(-l^2)|} \le 40K^2 e^{2K(\log(5/4)+\arctan 2)}.$$

Finally $\sum_{l \text{ odd}} |b_l| \le 4K^4$. Thus

$$|B(P,P)| \le 30720\pi K^7 e^{C_{1/2}K} G(P), \quad C_{1/2} = \pi/2 + 2\log(5/4) + 2\arctan 2 + 4\log(2+\sqrt{5}).$$

For any compressed positive matrix G and beta coefficient B_U, the symmetric operator $G^{-1/2} B_U G^{-1/2}$ has eigenvalues bounded in absolute value by this same quantity. If its nonzero eigenvalues are $\lambda_1, \dots, \lambda_h$, then

$$D_U(X) = D_U(\beta(4)) \prod_{i=1}^{h} (1 + (X - \beta(4))\lambda_i), \qquad \delta_U = \prod_{i=1}^{h} |\lambda_i|^{-1}.$$

This proves $\deg D_U = \operatorname{rank} B_U = h$, rather than assuming full rank. For $h = 0$ the normalized value is one. For $h \le K/5$ the operator bound gives (6.17). Its strict separation from (6.3) can be checked without decimals: $C_{1/2}/5 < 289/140$ using $\pi < 22/7$, $\log(5/4) < 1/4$, $\arctan 2 < 9/8$, $\log(2+\sqrt{5}) < 3/2$, while $(197 + 192\log 2)/150 > 13/6$ using $\log 2 > 2/3$.

C Evaluation in repeated-exponent spaces

This is the repeated-exponent form of Müntz–Laplace orthogonalization; see [1, Section 2] for the underlying orthogonal-system method. We give the needed norm estimate explicitly. For $\lambda > 0$, define

$$T_\lambda f(u) = f(u) - 2\lambda \int_0^u e^{-\lambda(u-v)} f(v) \, \mathrm{d}v.$$

On decaying exponential polynomials it is an $L^2(0,\infty)$ isometry. Indeed, with $h = e^{-\lambda u} * f$, the two functions are $h' + \lambda h$ and $h' - \lambda h$; their squared-norm difference is $-2\lambda[|h|^2]_0^\infty = 0$. Its range is orthogonal to $e^{-\lambda u}$ by direct integration.

For positive rates $\lambda_1, \dots, \lambda_N$, allowing repetitions, the functions

$$e_k = T_{\lambda_1} \cdots T_{\lambda_{k-1}} (\sqrt{2\lambda_k} e^{-\lambda_k u})$$

are orthonormal. Their Laplace transforms are $\sqrt{2\lambda_k}(s+\lambda_k)^{-1}\prod_{i<k}(s-\lambda_i)/(s+\lambda_i)$, so they belong to the exponential-polynomial space with these rates and multiplicities. Dimension then makes them a basis. Put $\Lambda = \max \lambda_i$ and $\sigma = \sum \lambda_i$. Expanding the finite convolution product at a complex point w, each term with m convolutions costs at most $|w|^m e^{\Lambda \max(0,-\Re w)}/m!$. Its scalar coefficients sum to at most $(2\sigma)^m/m!$. Since $\sum_{m\ge 0} a^m/(m!)^2 \le e^{2\sqrt{a}}$, Cauchy–Schwarz gives

$$|H(w)|^2 \le 2\sigma e^{2\Lambda \max(0,-\Re w)+4\sqrt{2\sigma|w|}}\|H\|_2^2. \tag{C.1}$$

For Section 7, take $H(u) = e^{-u/2}F(1+u)$. There are thirteen copies of each rate $2j+1/2$, $0 \le j < r$. Thus $\Lambda < 2r$ and $\sigma = 13r^2 - 13r/2 \le 13r^2$. Restriction of the original norm to $y \ge 1$ gives $\|H\|_2^2 \le 2e\|F\|_\mu^2$. At $w_l = z_l - 1$ the return to $F(z_l)$ contributes the squared factor e^{-1}. In (C.1), $\Re w_l = -1$ and $|w_l| \le M_0$. Consequently $|F(z_l)|^2 \le 52r^2 e^{(4+4\sqrt{26M_0})r}\|F\|_\mu^2$, which is the estimate used in Theorem 7.1.

D Quantitative inputs for the eleven linear forms

This appendix proves Lemma 8.2. We distinguish the coefficient-column clearer used for lattice height from the full clearer used for the resulting linear forms.

D.1 Local counts and the arithmetic envelope

At the pole $t = -a - 1/2$, let $c_{i,a}$ be the principal-part coefficients of $P(2t+n+1)R_n(t)$. For odd p with $p^2 > 7n$, define

$$F_p(a) = \sum_{j=0}^{28} \#\{v : d_j \le v \le n - d_j,\ v \equiv a \pmod p\} - 2\#\{\ell : -2n-1 \le \ell \le n,\ p \mid 2a+1+2\ell\}.$$

Taylor coefficient extraction gives $v_p(c_{i,a}) \ge i - F_p(a)$: each nonunit numerator factor supplies a power of p, every nonunit denominator factor costs one, and each extracted Taylor degree can cost at most one more. All nonzero differences have size below $7n < p^2$. The integer multiplier has integral local coefficients and cannot worsen the estimate. Isolated pole classes have integral principal parts without any extraction loss.

If M_p is the maximum of F_p over repeated classes, $\max(0, M_p+1)$ clears the complete form. For the rational constant, group the finite tails at the negative integers $-\ell$. The identity $(PR_n)'(-\ell) = 0$ permits either side of $-\ell$ to represent the same group. If the nonunit class is isolated choose the other side; otherwise a differentiated evaluation costs $i+1$ powers, leaving valuation at least $-F_p - 1$. In particular no odd prime $p > n$ is needed. For odd $p^2 \le 7n$, put $\lambda_p = \lfloor \log_p(7n) \rfloor$. At level p^k, the excess of denominator factors over numerator factors is at most $(16n+25)/p^k + 31$: the twenty-nine pole blocks contribute $22n+29$ terms, and the double zero interval removes $6n+4$. Consequently the constant local coefficient costs at most

$$\sum_{k=1}^{\lambda_p}\left(\frac{16n+25}{p^k} + 31\right) \le 16v_p(n!) + 72\lambda_p.$$

Taylor extraction adds at most $28\lambda_p$, and the finite-tail evaluation adds at most $30\lambda_p$. The sufficient exponent is

$$16v_p(n!) + 130\lfloor \log_p(7n) \rfloor.$$

These excesses sum to $o(n)$. At two, the squared half-integer numerator factors cost $6n+4$, the other pole factors at most $22n+29+29\lfloor\log_2 n\rfloor$, and extraction at most $28\lfloor\log_2 n\rfloor$. Therefore $16v_2(n!)+12n+80(\lfloor\log_2 n\rfloor+1)$ suffices.

Put $\delta_j = j/116$. For $x \geq 1$, use $0 \leq y \leq x-1$ when $x < 2$, and $0 \leq y \leq 1$ otherwise. Define

$$\begin{aligned} N_j(x,y) &= \#\{h \in \mathbb{Z} : \delta_j x \leq y+h \leq (1-\delta_j)x\}, \\ C(x,y) &= \lfloor y+x+1/2\rfloor - \lfloor y-2x+1/2\rfloor, \\ M(x) &= \max_y \left(\sum_j N_j(x,y) - 2C(x,y)\right), \\ e(x) &= \max(0, M(x)+1) - 16\lfloor x\rfloor. \end{aligned}$$

This majorizes the excess prime exponent at $x = n/p$. The elementary counts give $e(x) \leq 32+16\{x\}$. For $x \geq 2$, increasing x by 116 increases the pole count by 2552 and twice the zero count by 696. Since $M(x) \geq 16x-31 \geq 1$, the positive part causes no change and $e(x+116) = e(x)$.

The ordering, modulo one, of the finitely many affine endpoints $\delta_j x, (1-\delta_j)x, -x-1/2, 2x-1/2$ changes only at rational pairwise crossings or integer crossings. On each intervening open interval the envelope is constant. The supplied certificate enumerates all such cells: 125 in $[1,2]$ and 12966 in $[2,118]$. Summing thirty periods with upward rational rounding and bounding the tail after 3482 by $32/3482 + 8/3481$ gives

$$\int_1^\infty e(x)x^{-2}\,\mathrm{d}x \leq A = \frac{3669977723639606858759701275519 9}{3663299712317355278800988930048}. \tag{D.1}$$

The checks enumerate the defining endpoints and maxima, not just selected numerical values. For the asymptotic prime sum, apply the prime number theorem first on a fixed partition. Event points cost $O(\log n)$; the remote tail is controlled by the displayed majorant and Chebyshev's bound before the cutoff tends to infinity.

D.2 Cancellation improves the full clearer

For a repeated pole class, choose its least address $0 \leq a < p$. Put $x = n/p$, $r = a/p$ and $k = \lfloor x-r\rfloor$, so its addresses are $a, a+p, \ldots, a+kp$. When $x < 2$, repetition requires $r \leq x-1$. Define the fixed local rational function

$$H_{x,r}(u) = \frac{\prod_{h\in\mathbb{Z}+1/2,\ r-2x\leq h\leq r+x}(u-h)^2}{\prod_{l=0}^{k}(u+l)^{b_l}}, \qquad b_l = \#\{j : \delta_j x \leq r+l \leq (1-\delta_j)x\},$$

and its net order $F = \sum b_l - 2\#\{h\}$. After scaling, the actual half-integer numerator roots lie in

$$[r-2x-1/(2p),\ r+x+1/(2p)] \cap (\mathbb{Z}+1/2).$$

This contains the ideal interval used in $H_{x,r}$. Each additional root supplies p^2, so it can only remove a surviving local term when the intended improvement uses at most two powers of p. For a proper rational function T with these poles and principal parts T_l, put

$$\mathcal{B}(T) = \sum_{l=0}^{k}(-1)^l \operatorname{res}_{-l} T, \qquad \mathcal{Q}(T) = \sum_{v=0}^{k-1}(-1)^v \sum_{l=v+1}^{k} T_l'(-v-1/2).$$

On a cell where $M = \max F$, suppose $g \in \{1,2\}$, $M \geq g+1$, and one rational κ satisfies

$$\mathcal{Q}(u^h H_{x,r}) = \kappa\mathcal{B}(u^h H_{x,r}) \quad \text{for all } F \geq M-g+1, \quad 0 \leq h < g-(M-F). \tag{D.2}$$

Then g powers of p can be removed from the old clearing exponent for every beta-two-canceling multiplier, apart from the fixed primes in the denominator of κ.

Here is the local-to-global justification. Substitute $t = -a - 1/2 + pu$ and multiply by p^M. Unit-factor inverses have Taylor coefficients divisible by the appropriate powers of p. Extra numerator zeros cost p^2. Modulo p^g, the surviving terms are therefore

$$\sum_{0 \le h < g-(M-F)} c_{a,h} p^{M-F+h} u^h H_{x,r}(u), \qquad c_{a,h} \in \mathbb{Z}_{(p)}.$$

For bounded x, all local pole differences and half-integer evaluation distances are units when p is large. Principal-part extraction commutes with this reduction. The beta-two coefficient is -4 times the sum of the signed local $\mathcal{B}$ values, scaled by p^{1-M}. The rational constant has the exact grouped formula

$$q(P) = -\sum_{\ell=1}^{n} (-1)^\ell \sum_{a \ge \ell} f_a'(-\ell), \qquad f = PR_n,$$

and either side of $-\ell$ may be used. The nonunit evaluation between addresses $a + vp, a + (v+1)p$ is at $\ell = a + (p+1)/2 + vp$, corresponding to $u = -v - 1/2$. It gives, up to a common sign, the signed sum of local $\mathcal{Q}$ values scaled by p^{-M-1}. Unit-denominator terms have at least two powers after this scaling. Thus (D.2) and exact beta-two cancellation give g powers in the cleared rational constant. The remaining beta coefficients, of indices at least four, already have enough powers. This proves the claimed improvement for the same forms, including their rational constants.

For example, in $n/2 < p < 15n/29$ the maximal local function is

$$H(u) = \frac{\prod_{h=-5/2,-3/2,-1/2,1/2,3/2} (u-h)^2}{u^{29}(u+1)^{29}}.$$

Reflection makes both functionals vanish on $H(u)(u+1/2)$. Exact evaluation gives $\mathcal{Q}(H)/\mathcal{B}(H) = -15010513046/139778381$, so $g = 2$ works outside its fixed denominator primes. The broader certificate checks (D.2) on every relevant pattern in 582 open cells of $[1, 6]$ and obtains

$$S = \int_1^6 g(x) x^{-2} \, \mathrm{d}x = \frac{62795}{199056}. \tag{D.3}$$

Event points and fixed exceptional primes receive no saving. It follows that a common full clearer satisfies

$$\log D_n^* \le 16 \log(n!) + (A + 12 \log 2 - S + o(1))n. \tag{D.4}$$

D.3 Uniform analytic decay

Let $H(v) = v \log v$ for $v > 0$, with $H(0) = 0$, and for $u \ge 1$ put

$$\Phi(u) = 2[H(u+2) - H(u-1)] - \sum_{j=0}^{28} [H(u+1-\delta_j) - H(u+\delta_j)].$$

The exact gamma product for R_n gives, uniformly for $t \ge n+1$,

$$\log R_n(t) = -16 \log(n!) + n\Phi(t/n) + O(\log(n+t)).$$

Uniformity includes $t = n+1$: the smallest gamma argument is one, and $\log \Gamma(v) = v \log v - v + O(\log(v+2))$ is uniform for $v \ge 1$. The finite reciprocal-factor sum bounds the logarithmic derivative

by $O(1+\log(n+t))$. Moreover the multiplier and its derivative are bounded by $Ch_n(P)(1+t/n)^{30}$ and $Ch_n(P)n^{-1}(1+t/n)^{29}$.

The phase has a unique maximum near 1.00165. One verifies $\Phi'' < 0$ on $(1, 11/10]$ by the rational upper bound

$$2/3 - 20 + \sum_j ((1+\delta_j)^{-1} - (21/10 - \delta_j)^{-1}) < 0.$$

For $u \geq 11/10$, $\Phi'(u) \leq 2\log((u+2)/(u-1)) - 22/(u+1) < 0$: the negative of this bound has no interior minimum, is positive at $11/10$, and tends to zero. For $u \geq 4$, the stronger $u\Phi'(u) < -9$ controls the infinite tail. Rational log enclosures and a bracket for the critical point give the supplied upper bound

$$\Phi_* \leq -\frac{4093415547496757252324770508326269837463}{17014118346046923173168730371588410572 8} = -24.0589342582\ldots.$$

The checker independently encloses the maximum with a tangent on the concave interval. All summands through $t = n$ vanish; the tail bound absorbs the multiplier's fixed degree. Thus

$$|\mathcal{J}_n(P)| \leq h_n(P)(n!)^{-16}e^{(\Phi_*+o(1))n}.$$

Combining with (D.4) proves the second part of Lemma 8.2, with $C_* = -A - 12\log 2 - \Phi_* + S > 6.0384$.

D.4 The cheaper column clearer

The beta-two coefficient is $-4\sum_a (-1)^a c_{1,a}$. At odd large primes it needs only $\max(0, M_p - 1)$ instead of $\max(0, M_p + 1)$. The envelope saving equals two for $x \geq 2$. On $[1, 2]$ the exact partition gives integral $107/116$, so the total saving is $223/116$. At two, put $z = t + 1/2$. Multiplication by 2^{6n+4} makes the numerator integral. The even-address and odd-address denominator products reduce to powers of z and $z+1$. Their resultant is odd, so partial fractions by these two groups have numerators in $\mathbb{Z}_{(2)}[z]$. Each group's residue sum is the coefficient of z^{-1} at infinity and is integral at two. The total residue sum is zero by proper decay, so the alternating sum is twice one group sum. Consequently the beta-two coefficients have valuation at least $-6n - 1$, uniformly in the multiplier.

A column clearer E_n therefore satisfies

$$\log E_n \leq 16\log(n!) + (A - 223/116 - 10\log 2 + o(1))n.$$

The coefficient estimate uses a different profile from the positive tail phase Φ. Put $\mathcal{H}(v) = v\log|v|$ for $v \neq 0$, $\mathcal{H}(0) = 0$, and define, for $0 \leq x \leq 1$,

$$\Psi(x) = 2[\mathcal{H}(1+x) + \mathcal{H}(2-x)] - \sum_{j=0}^{28}[\mathcal{H}(1-\delta_j - x) + \mathcal{H}(x - \delta_j)].$$

On the radius-$1/4$ circle about the pole $-a - 1/2$, split each product at its closest integer. Summing changes $\log(1 + O(1/|k|))$ costs $O(\log n)$ per block. Stirling's formula therefore gives, uniformly in the address a including endpoints,

$$\log|R_n(t)| = -16\log(n!) + n\Psi(a/n) + O(\log n).$$

The profile is symmetric about $1/2$. Away from its finitely many cusp points,

$$\Psi'(x) = 2\log\frac{1+x}{2-x} + \sum_{j=0}^{28}\log\frac{|1-\delta_j - x|}{|x - \delta_j|}.$$

On $0 < x < 1/2$, every $j > 0$ term is positive, and the $j = 0$ term dominates the negative numerator contribution because $(1-x)(1+x)^2 - x(2-x)^2 = (1-2x)(x^2-x+1) > 0$. Continuity across the cusps gives the midpoint maximum

$$\Psi_* = \Psi(1/2) = 6\log(3/2) - 2\sum_{j=0}^{28}(1/2-\delta_j)\log(1/2-\delta_j).$$

Cauchy's residue formula then bounds the coefficient column by $(n!)^{-16}e^{(\Psi_*+o(1))n}$. Clearing and removing its integer gcd can only improve this estimate. Thus

$$H_* = A - 223/116 - 10\log 2 + \Psi_* = 24.5217727858\ldots < 24.522,$$

proving the first part of Lemma 8.2.

E Record coverage and reproducibility

The accompanying guide maps the manuscript to its supporting records and identifies which checks are recomputations, certificate replays, or retained finite observations. The mathematical claims are separated as follows.

Construction	Conclusion and remaining condition
Monic zeta substitution	Exact all-subset valuation; the globally cleared q^{-2} certificate is excluded.
Full-radical compression	Integral period elimination with $o(K^2)$ relative-height cost; compatible determinant arithmetic and decay open.
Low-degree compression	Uniform prime and dyadic content bounds; true primitive cost versus achieved decay unresolved.
Thirteen-value Gram system	Exact rational-block removal and linear-scale residual decay; primitive arithmetic unresolved.
Fourteen-value linear forms	Formal rank, classified identities, and eleven uniformly small independent rows; collective nonvanishing open.
Fixed-direction transport	Cannot shrink a complete collection while preserving each individual rational coefficient line.
Loewner specialization	Nonvanishing hypotheses checked; one compatible coarse cost comparison inconclusive.

The final content bounds supersede the weaker intermediate allowances; the latter are not separate factors to be multiplied into the result. Finite tables are retained to document the evidence that motivated the general questions, not as substitutes for their missing limits.

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